

Automatic mode-locked fiber laser based on adaptive genetic algorithm

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ARTICLE INFO

Keywords:

Adaptive genetic algorithm
Stretched pulse
Passively mode-locked fiber laser

ABSTRACT

A modified adaptive genetic algorithm (GA) is proposed and implemented in a mode-locked erbium-doped fiber laser (EDFL) based on nonlinear polarization rotation. The algorithm combines GAs with adaptive crossover and mutation rates. As the number of iterations increases during the solving operation, the crossover and mutation rates are dynamically adjusted with the change in the fitness function value. A stretched pulse was automatically generated when the modified adaptive GA was continuously optimized to converge the intracavity polarization state to the target state. The average number of generations required to adjust from other states to stretched pulse locking is two, which is better than that afforded by the general GA. The evolution curves of the mode-locked EDFL with different crossover and mutation rates were investigated, and the results indicate the effectiveness of the proposed modified adaptive GA. This study proposes a new scheme for rapid polarization searching in automatic mode-locked fiber lasers.

1. Introduction

In recently years, ultrashort pulses have been used in many fields (e.g., optical fiber communication, medicine, optical imaging, and precision instrument manufacturing) because of their high peak power and narrow pulse width [1–4]. Passive mode-locked fiber laser (PMLFL) technology is crucial for generating ultrashort pulses [5,6]. PMLFL technology can be classified into nonlinear polarization rotation (NPR) technology, nonlinear optical loop mirror technology, semiconductor saturable absorber mirror, single-walled carbon nanotubes, and graphene [3,7–9]. Among these technologies, NPR is preferred because of its simple structure, high damage threshold, and narrow output pulse width [10]. However, the operating state of NPR-based PMLFL is sensitive to environmental temperatures and mechanical vibrations. Therefore, conventional NPR-based mode locking is restricted in terms of stability improvement.

Owing to the advent of artificial intelligence (AI) technology, AI algorithms have been used extensively in ultrafast photonics [11]. Among them, machine learning contributes significantly to investigations pertaining to automatic mode locking algorithms of MLFLs owing to its outstanding computing performance and efficient processing of large data sets [12–16]. Hellwig et al. used a piezoelectric squeezer as an automatic polarization controller to regulate the polarization state. They

obtained the distribution of clamping features in different polarization states using NPR technology [17], thus providing a solid foundation for automatic mode locking technology. Kutz et al. combined machine learning and automatic mode-locking technology and proposed the concept of “self-tuning fiber laser” [18]. R. Woodward et al. proposed the use of genetic algorithms (GAs) to optimize the polarization state in a cavity to achieve automatic mode locking. However, owing to the delay in the remote interface between instruments, the algorithm optimization process requires approximately 20 generations on average [19]. Pu et al. proposed an advanced Rosenbrock searching algorithm for the automatic mode-locked control of an NPR fiber laser, which significantly improved the responsiveness of the algorithm to the laser [20].

A modified GA with adaptive crossover and mutation rates was used to automate the self-tuning of an erbium-doped fiber laser (EDFL). The crossover and mutation rates could be adjusted accordingly during the operation of the algorithm to improve the mode locking efficiency. In the experiment, a modified adaptive GA was used to control the data acquisition (DAQ) card, and the output DC control voltage directly affected the electric polarization controller (EPC). Accordingly, a stretched pulse with a 3-dB spectral bandwidth (BW) of ~ 31 nm and a pulse duration of 2.22 ps was obtained. The average number of mode-locked generations required to adjust from other states to stretched pulse locking was two. The evolution curves of mode-locking with

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<https://doi.org/10.1016/j.yofte.2024.103677>

Received 23 October 2023; Received in revised form 11 December 2023; Accepted 10 January 2024

Available online 16 January 2024

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different crossover and mutation rates under the control of the GA were investigated.

2. Experimental setup

Fig. 1 illustrates the core adaptive genetic algorithm concept, including the fitness calculations, selection, individual assessment, crossover rate, variation rate adjustments, crossover, and variation. The initial population has a great influence on the performance of the GA [21]. The “goodness” of the initial population is affected by the diversity in the population, the performance of the algorithm is not favorable if the initial population is set too large or too small [22]. Empirically, the initial population was set to 20. First, an initial population of 20 individuals (i.e., zero generation) was randomly generated. Each individual comprised four genes that corresponded to the four channels of the DC control voltages for the EPC. In this algorithm, each person is evaluated using a fitness function and assigned a score. The selection of the fitness function is crucial for promptly and accurately distinguishing between operation regimes. A dual-region count (DRC) scheme was adopted to preliminarily judge the waveform data output by an

oscilloscope [20]. When the obtained waveform data satisfied the DRC, the waveform was classified into two regions: pulses and noise. To facilitate pulse counting, two thresholds, i.e., T_1 and T_2 ($T_1 > T_2$), were specified. In the pulse region, when the waveform intensity (denoted as P) $P > T_1$, the pulse was counted and the region was considered as pulse region. When $P < T_2$, the region was considered noisy. Ideally, $P > T_2$ should be absent in the noise regions. Consequently, the pulse count N in the pulse region should satisfy $N = \text{floor}(N_1/N_2)$, where the *floor* function is rounded down, N_1 represents the number of sampling points per period of the oscilloscope, and N_2 indicates the number of points between two pulses at the fundamental repetition frequency. Mode-locked pulses have considerably larger amplitudes than free-running pulses. In this study, the fitness function is expressed as shown in Eq. (1).

$$\text{fitness} = \frac{1}{N} \sum_{i=1}^N P_i \quad (1)$$

where P_i denotes the intensity of the pulse waveform and N is the pulse count when the DRC is satisfied. Therefore, the fitness function is the average of the counted pulse amplitudes.

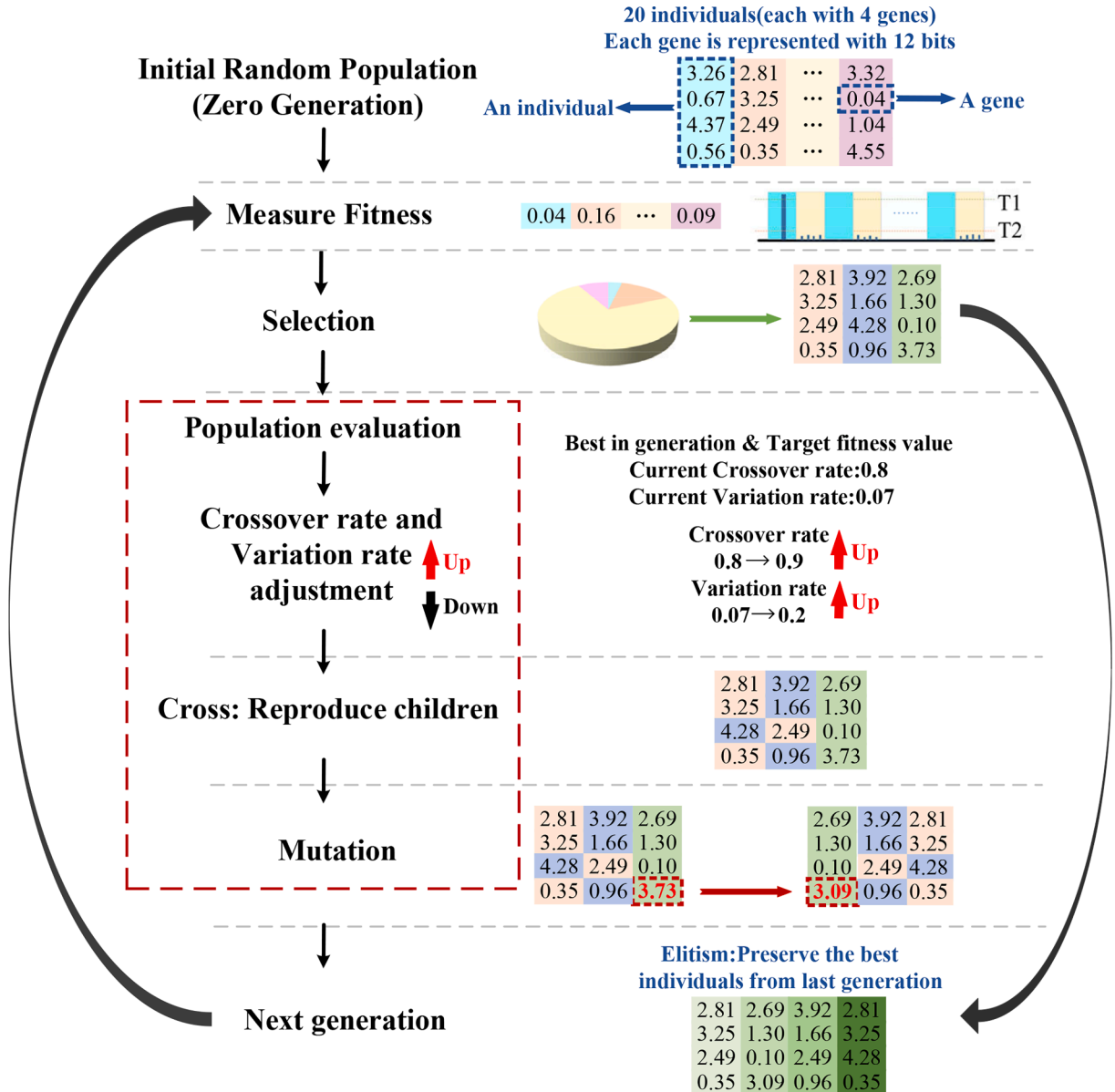


Fig. 1. Schematic illustration of adaptive genetic algorithm.

In GAs, the roulette wheel selection is used to select individuals from the previous generation to generate the next generation, where the probability of selecting each individual is directly proportional to its fitness [23]. The algorithm starts with 20 random individuals and then selects four of them as the population to generate the 1st generation. Among the four individuals, three are chosen using ‘roulette wheel’ selection, selecting three individuals with high fitness. The fourth individual is the highest fitness individual among these 20 random individuals. The crossover rate refers to the probability of recombination or exchange of genetic material between two parent individuals during the reproduction process in a genetic algorithm. The mutation rate is the probability of introducing random changes or alterations in the genetic information of an individual during reproduction [24,25]. Crossover is a primary method for generating new individuals: a higher crossover rate increases the diversity of the population and provides a greater probability of obtaining the global optimal solution. A higher mutation rate maintains the population diversity and prevents the algorithm from becoming stuck in the local optima. Meanwhile, lower crossover and mutation rates contribute to the overall stability of the population [26,27]. During the mode locking process, the time required for the algorithm to converge to the target state must be accelerated to the maximum extent. Fixed crossover and mutation rates do not satisfy these requirements. Therefore, an adaptive crossover and mutation rate were designed for the GA. For the adaptive strategy, the adaptive crossover (p_c) and mutation (p_m) probabilities are calculated based on the current population maximum fitness f_{max} , the average fitness f_{avg} , and the fitness of each individual f_i . The formulas are as follows [28]:

$$p_c = \begin{cases} k_1 & f_c \leq f_{avg} \\ k_1 - \frac{(k_1 - k_2)(f_c - f_{avg})}{f_{max} - f_{avg}} & f_c > f_{avg} \end{cases} \quad (2)$$

$$p_m = \begin{cases} k_3 & f_m \leq f_{avg} \\ k_3 - \frac{(k_3 - k_4)(f_m - f_{avg})}{f_{max} - f_{avg}} & f_m > f_{avg} \end{cases} \quad (3)$$

where f_c represents the larger fitness values among the solutions to be crossed. k_1, k_2 are the maximum and minimum values of the crossover probability, respectively. f_m denotes the larger fitness value of the solutions to be mutated, k_3, k_4 are the maximum and minimum values of the mutation probability, respectively. From equations (2) and (3), it can be observed that, the values of crossover and mutation rates are adjusted based on the relationship between the fitness of the current individual and the average fitness of the population. Once three individuals were selected via the roulette wheel selection, the optimal fitness of the current population was evaluated. The crossover rate varied between 0.8 and 0.9 and the mutation rate varied between 0.07 and 0.2, depending on the evaluation results. In particular, the mutated genes must be within 0–5. When evaluating the best individual in a population, a lower fitness score indicates a greater distance from the desired state, thus implying the necessity for higher crossover and mutation rates. As the population increases, the crossover and mutation decrease gradually to avoid disrupting excellent individuals when individuals with higher fitness levels (close to the target pulse) appear gradually in the population. A reinsertion operation is performed to generate the next generation and preserve the best individuals from the previous generation. When a new population consisting of four individuals is formed, it is considered that the population has completed one iteration process. After completing one iteration process, the new population is subjected to a fitness function. In the algorithm, it is defined that if more than 80 % of the threshold for mode locking is exceeded, it is determined as mode-locked state. If the score of each individual in the population remains below 80 % of the mode-locking threshold, it is deemed that the population has not reached the mode-locked state. Consequently, the algorithm successfully moved on to the next round of iteration.

Fig. 2 shows a schematic illustration of the intelligent MLFL, in which a 980 nm laser was used as the pump source and the pump light was coupled to the mode-locked resonant cavity through a 980 nm/1550 nm wavelength-division multiplexer. A section of EDF with the length of 13.5 m was used as the gain medium. Additionally, a 20:80 optical coupler was used to maintain 80 % of the power inside the cavity for oscillation, whereas the remaining 20 % was the output for detection. The polarization sensitive isolator (PS-ISO) guaranteed the unidirectional operation of the cavity and functioned as a polarizer. As the core component of automatic mode locking, the EPC was driven by four channels with a 0–5 V DC voltage. The other segments were single-mode fibers (SMFs) with a total length of 9.9 m. The dispersion coefficients of the EDFs and SMFs were -9 and 17 ps/nm/km, respectively. Hence, the net dispersion of the cavity was approximately -0.06 ps², which signifies near-zero dispersion and satisfies the condition of stretched pulse.

The output signal was detected using a high-speed photodetector, obtained using the oscilloscope, and transferred to the computer. The status of the laser was determined using a GA. Subsequently, the computer sent control instructions to the DAQ card in each control cycle, which served as the digital-to-analog converter for the EPC. The DAQ card synchronously output four 0–5 V DC control voltages with micro-second response times to the EPC in accordance to the computer instructions. These steps were repeated individually until the desired regime was achieved. In the experiment, the output laser was measured in real time using a spectrum analyzer (Yokogawa AQ6370D) and an autocorrelator (APE Pulsecheck USB150).

3. Experimental results and discussion

In the experiment, a stable stretched pulse was achieved at 200 mW of pump power. The typical output spectrum is shown in Fig. 3(a) with a central wavelength of ~ 1560 nm, where the spectral curve conformed to a Gaussian function and was smooth without Kelly sidebands. The 3-dB spectral BW is approximately 31 nm. The corresponding autocorrelation signals of the experimental data and Gaussian fit are shown in Fig. 3(b). The pulse durations are approximately 2.22 ps. Therefore, we calculated the time-bandwidth product (TBP) of this stretched pulse to be approximately 8.59, which indicates the appearance of a large chirp on the output pulse. Maintaining the pump power and properly adjusting the PC state, the spectral width of the stretched pulse ranges from 18.14 nm to 32.97 nm, while the pulse duration varies between 2.2 ps and 3.48 ps. The reason for this variation lies in the NPR-related filtering effect [29]. Fig. 3(c) shows the oscilloscopic signals of the pulse train. The intensities of the pulses are.

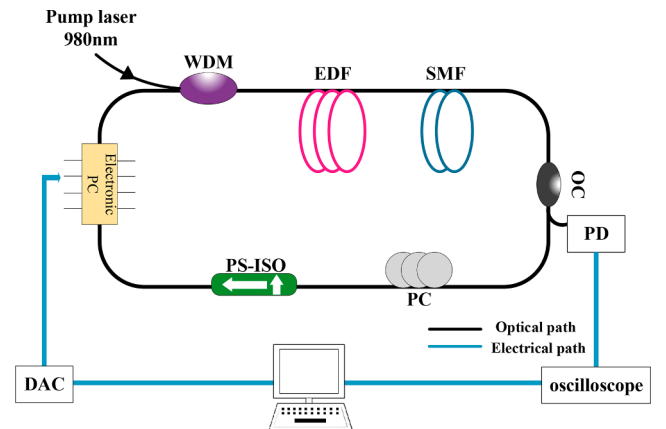


Fig. 2. Schematic illustration of intelligent erbium-doped mode-locked fiber laser. WDM: wavelength-division multiplexer; EDF: erbium-doped fiber; SMF: single-mode-fiber; OC: optical coupler; PD: photodetector; PC: polarization controller; PS-ISO: polarization sensitive isolator; EPC: electric polarization controller.

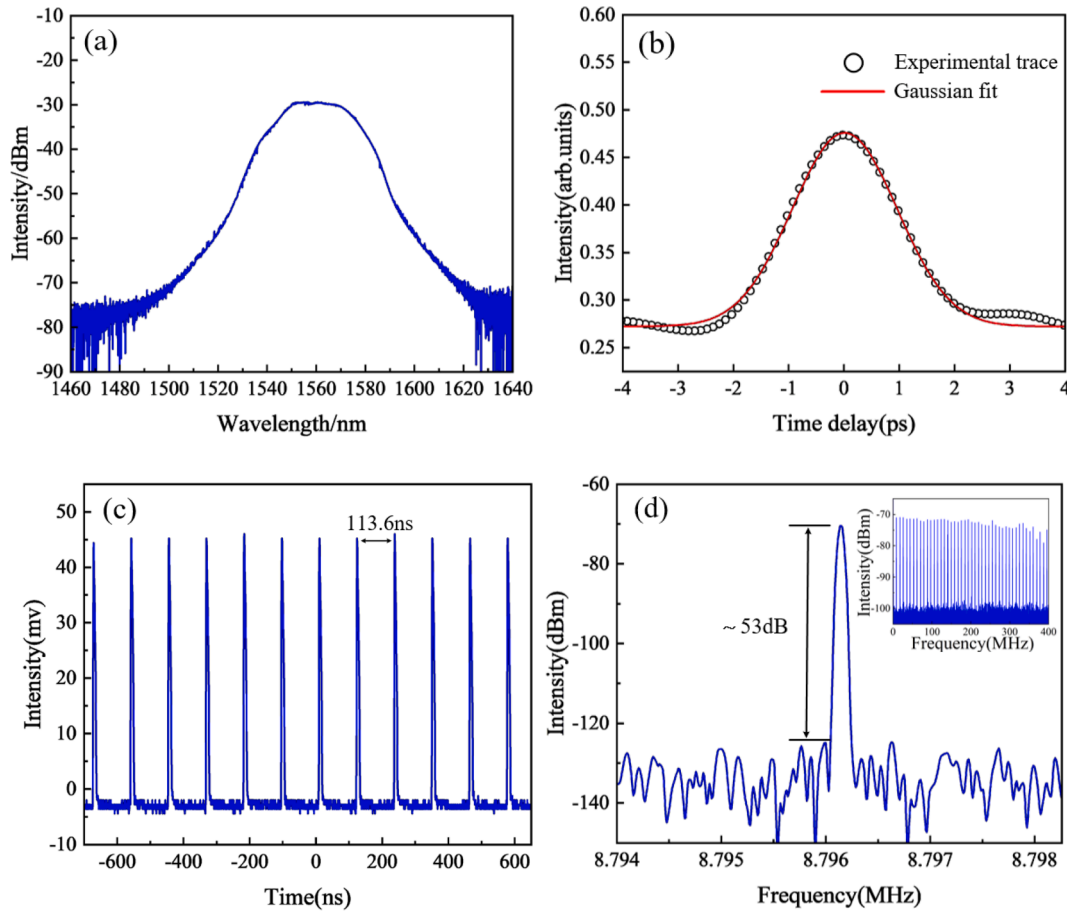


Fig. 3. Output of typical stretched pulse at 200 mW of pump power. (a) Optical spectrum; (b) autocorrelation signals; (c) oscilloscope signals; (d) RF spectrum of fundamental cavity repetition rate. Inset shows RF spectrum with 400 MHz span.

almost equal and evenly distributed. The time interval between the two pulses is ~ 113.6 ns, which corresponded to a cavity length of 23.4 m. Fig. 3(d) shows the corresponding radio-frequency (RF) spectrum of the mode-locked pulse. The fundamental repetition frequency of the pulse train is approximately 8.796 MHz. The signal-to-noise ratio is approximately 53 dB. The inset of Fig. 3(d) shows the 400 MHz-range spectrum curve. No modulation is observed, which indicates that the laser is in a continuous mode-locked state without Q-switching instability.

In the experiment, 80 tests were conducted, as shown in Fig. 4(a). The average number of generations required to adjust from other states to stretched pulse locking is two. Based on the data shown in the chart, 80 sets of test results were achieved within nine generations. Fig. 4(b) presents a bar chart depicting the frequency of mode locking from the first to the ninth generation in Fig. 4(a). As shown, the second generation of mode locking constituted the highest proportion, i.e., 45 % of the total. Moreover, the sum of the mode locking probabilities within the four generations reached 94 %, thus reflecting the excellence of the

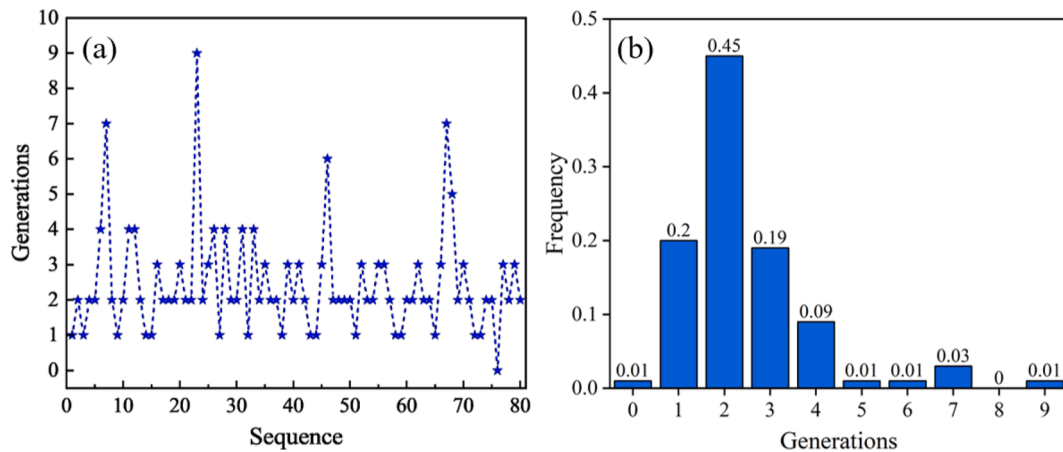


Fig. 4. (a) Generations corresponding to 80 mode-locked stretched pulses; (b) bar chart depicting frequency of mode locking from first generation to ninth generation.

proposed algorithm.

To validate the effectiveness of the modified adaptive GA, the evolutionary curves of the best and average fitness scores, as well as the proportion of mode-locked individuals in each generation, indicating the mode-locked probability in each generation, were analyzed, as shown in Fig. 5. The black diamond and blue square curves represent the average and optimal fitness values for each generation, respectively. it is assumed to be in a mode-locked state. Therefore, the laser was at lose mode-locked.

state when the population iterations were performed in the first two generations, which corresponded to low fitness values. As the population iterated to the third generation, the best fitness curve indicated a mode-locked state, which gradually stabilized as the generations increased. When the population reached the fourth generation, the average fitness curve indicated a mode-locked state. The red triangular curve represents the proportion of mode-locked individuals in each generation, indicating the mode-locked probability in each generation. In the third generation, the proportion of mode-locked individuals in the population was 50 %, thus indicating that mode locking began with the third individual in the population. Therefore, in the 3rd generation of the population, two individuals have not yet reached the mode-locked state, their fitness scores are relatively lower. When the average fitness score of the population is calculated, the result is below 80 % of the mode locking threshold. The result is that the evolution curve of the average fitness score is not mode-locked in the 3rd generation, which results in the evolution curves of the average and optimal fitness scores are not locked at the same generation. The three function curves in the graph clearly show a converging trend from low to high, followed by stabilization. This demonstrates the effectiveness and stability of the proposed algorithm.

To further validate the effectiveness of the proposed algorithm, the evolution curves of fitness with crossover and mutation rates were investigated. In the experiment, control variable method was employed. Here, the mutation rate was kept constant at 0.07. The effects of different crossover rates were investigated, as shown in Fig. 6. Individual assessments were conducted in the experiment based on the average of 10 fitness calculations. Based on the data in the graph, one can conclude that under the same mutation rate, different parameters were used for the crossover rate, including 0.2, 0.4, 0.6, 0.8, and 1.0. When the crossover rate was set to 0.2, a stretched pulse was observed in the eighth generation. Meanwhile, when the crossover rate was set to 0.4, mode locking was achieved in the sixth generation. For crossover rates of 0.6, 0.8, and 1.0, mode locking occurred in the fourth, third, and fourth generations, respectively. Under different crossover rates, the average fitness increased with the increase of iterations. In addition, as the crossover rate increased gradually, the mode-locking speed increased. However, when the crossover rate reached a certain level, the

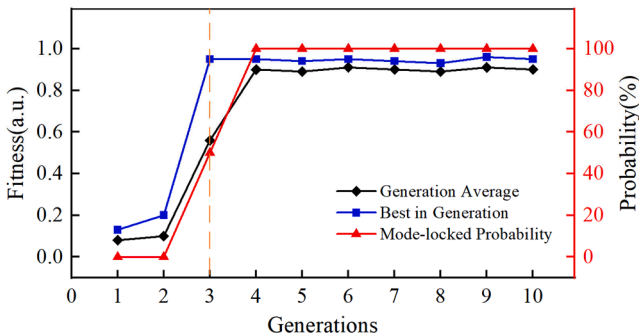


Fig. 5. Black diamond curve represents evolution of average fitness score for successive generations; blue square curve represents evolution of best fitness score for successive generations; and the red triangular curve represents the proportion of mode-locked individuals in each generation, indicating the mode-locked probability in each generation.

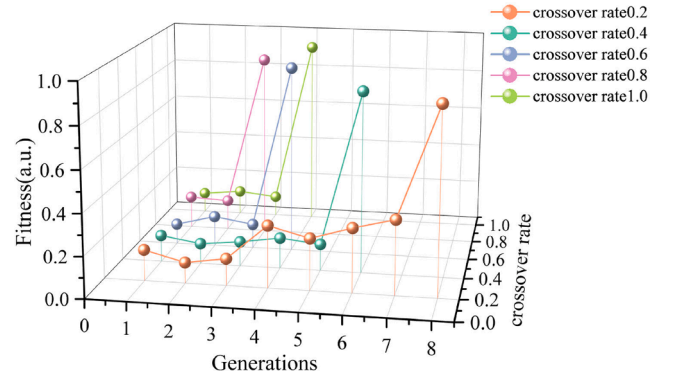


Fig. 6. Evolution curves of fitness at different crossover rates.

mode locking speed decreased. This is because when the crossover rate was low, less gene exchange occurred, thus resulting in a low genetic diversity of new-generation individuals and a relatively lower evolutionary speed. The higher the crossover rate, the greater was the genetic diversity of new-generation individuals and the faster was the evolutionary speed. When the crossover rate was extremely high, gene exchange between parent individuals became frequent, which affected the evolutionary speed.

The effects of different mutation rates were investigated, as shown in Fig. 7. Here, the crossover rate is maintained at 0.8. Individual assessments were conducted in the experiment based on the average of 10 fitness calculations. At the same crossover rate, the mutation rates were set to 0.01, 0.03, 0.05, 0.07, and 0.09. As shown in the graph, when the mutation rate was set to 0.01, mode locking occurred in the seventh generation. When the mutation rate was set to 0.03 and 0.09, mode locking occurred in the fifth generation. When the mutation rate was set to 0.05 and 0.07, a stretched pulse was achieved in the third generation. Under different mutation rates, as the iterations increased, the average fitness and mode locking speed exhibited a trend similar to the crossover rate. This is attributable to two main reasons: First, the mutation process occurred randomly and was uncontrollable during the experiment. Second, new-generation individuals were generated through the crossover and mutation of previous-generation individuals. When the mutation rate was low, new individuals were primarily affected by the crossover operations. As the mutation rate increased, the genetic diversity of the individuals increased as well, which facilitated the evolution of the population. However, when the mutation rate was higher, the algorithm introduced new genomes frequently, thus increasing the probability of disrupting better individuals from the previous generation and resulting in the slower convergence of the algorithm. Constrained by the range of the mode-locked region [30] and the rapid mode locking speed, the fitness evolution curve exhibits almost no transitional states.

Therefore, selecting the appropriate crossover and mutation rates is

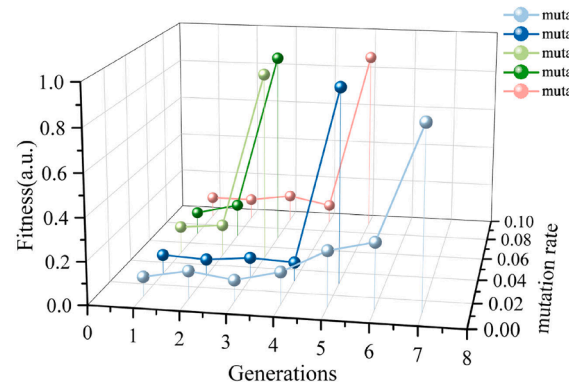


Fig. 7. Evolution curves of fitness at different mutation rates.

crucial a high performing GA. Based on the experimental results, for the GA with fixed crossover and mutation rates, the algorithm converges from other states to stretched pulse locking in a minimum of three generations. However, the modified adaptive GA required only two generations. Thus, the effectiveness of the modified adaptive GA was demonstrated.

4. Conclusion

We demonstrated an intelligent MLFL based on NPR technology by employing a modified adaptive GA. In the EDFL constructed, as shown in Fig. 2, a GA with adaptive crossover and mutation rates was used to control the internal dynamic process to generate ultrafast pulses. When the pump power is 200 mW, the laser output is stable stretched pulse mode locking with a 3-dB BW and a TBP of approximately 31 nm and 8.59, respectively. After 80 tests, the average number of generations required to adjust from other states to stretched pulse locking is discovered to be two. In addition, the sum of the probabilities of mode locking within the four generations reached 94 %. The evolution curves of mode locking with different crossover and mutation rates under the control of the GA were investigated, which demonstrated the effectiveness of the proposed modified adaptive GA. We believe that this efficient automatic MLFL will contribute significantly to the rapid polarization search of lasers.

Funding

This work was supported by the National Key R&D Program of China (2022YFB2903201), National Natural Science foundation of China (Nos. 61805198, 62105260, 12304373), Key R&D Project in Shaanxi Province of China (No. 2022GD-TSLD-50), Natural Science Basic Research Plan of Shaanxi Province of China (Nos. 2021JQ-720, 2022JM-357, No.2022JQ-638).

CRediT authorship contribution statement

Dongdong Han: Conceptualization, Writing – original draft, Formal analysis, Investigation, Methodology, Writing – review & editing. **Ruotong Guo:** Methodology, Validation, Writing – review & editing, Writing – original draft. **Guojun Li:** Formal analysis, Validation. **Yani Chen:** Data curation, Visualization. **Boyuan Zhang:** Software. **Kaili Ren:** Formal analysis, Funding acquisition. **Yipeng Zheng:** Formal analysis, Funding acquisition. **Lipeng Zhu:** Data curation, Formal analysis. **Tiantian Li:** Funding acquisition, Validation. **Zhanqiang Hui:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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